

CAMA: Implementation and Functional Evaluation of an Emotionally-Indexed

Semantic Memory Architecture for Human–AI Interaction Continuity

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Abstract

This paper presents the implementation and functional evaluation of the Circular Associative Memory Architecture (CAMA), a three-layer emotionally-indexed semantic memory system designed for longitudinal human–AI interaction. Building on prior theoretical work (Reinhold, 2026a; 2026b), we describe a fully functional implementation comprising 9,100+ stored memories with local semantic embeddings, a counterweight safety mechanism, and assistant-side persistent state tools enabling bidirectional memory storage. The system runs entirely on-device with no cloud dependency, using sentence-transformers for semantic retrieval. We report results from initial functional tests demonstrating that the retrieval and counterweight mechanisms activate as designed under high-negative-affect queries. We further introduce bidirectional memory—where both human-authored records and assistant-generated response records are stored within the same architecture—and outline implications for companion systems and future controlled evaluation. Findings are preliminary and based on a single-dyad implementation; multi-participant validation is planned as future work.

Keywords: AI memory architecture, emotional indexing, semantic retrieval, human-AI interaction, companion systems, counterweight safety, bidirectional memory, persistent state, MCP protocol

1. Introduction

Many current AI memory systems prioritize keyword-based or semantic retrieval within limited-capacity storage (typically 20–30 memory slots), with minimal emotional indexing. Commercial implementations from major AI providers generally treat memory as preference storage rather than experiential continuity infrastructure. The Circular Associative Memory Architecture (CAMA) was proposed as an alternative: a three-layer system in which memories carry multi-dimensional emotional annotations, retrieval is blended across semantic, affective, relational, and temporal dimensions, and write discipline distinguishes between user-authored durable teachings and AI-generated provisional inferences.

This paper reports on the first complete implementation of CAMA, deployed on a consumer desktop machine via the Model Context Protocol (MCP). We present the system architecture, implementation details, initial functional test results, and two contributions: (1) a counterweight safety mechanism designed to prevent retrieval-induced emotional spiraling, and (2) a bidirectional memory layer where assistant-generated response records are stored and retrieved alongside human-authored records.

2. System Architecture

2.1 Three-Layer Design

CAMA implements three layers: **Shelves (Archive)**—immutable raw text with recomputable emotional annotations and semantic embeddings, providing unlimited durable storage. **Racks (Relational Index)**—edges between memories typed by relationship: resonance, contradiction, elaboration, deepens, transforms, echoes. **Console (Active Ring)**—a circular buffer of 30 slots representing working memory, with least-recently-activated eviction.

2.2 Write Discipline

A core design principle is the distinction between user-authored teachings and AI-generated inferences, each governed by different persistence and weight rules. Table 1 summarizes the write discipline.

Source	Status	Weight	Expiry	Confirmation
Teaching (user)	durable	100%	None	Not needed
Inference (AI)	provisional	40%	TTL (7d default)	Required
Expired inference	expired	0%	—	<i>Not confirmed ≠ contradicted</i>
Contradicted	rejected	0%	—	Kept for audit

Table 1: Write discipline—teachings vs. inferences.

2.3 Blended Retrieval

Retrieval scores are computed as a weighted blend of four signals: semantic similarity (0.45) via cosine distance on embedding vectors, affect resonance (0.25) via hybrid valence/arousal distance combined with discrete emotional chord matching, relational weight (0.15) via precomputed edge degree, and recency decay (0.15) with a 30-day half-life. Core memories receive a score boost. Results are drawn from a shortlist of 500 candidates to balance accuracy with computational cost. These weights were selected through iterative design rather than systematic optimization; ablation studies varying weight distributions are a priority for future work.

2.4 Anti-Spiral Counterweight System

When the system detects strongly negative affect in the current query context (defined as aggregate negative emotional scores exceeding a configurable threshold), an anti-spiral mechanism activates. The system injects counterweight memories drawn from five typed categories: grounding (anchoring to what is stable), agency (evidence of capability and accomplishment), connection (evidence of relationship and support), self-compassion (recognition of one's own path), and evidence of progress (documented achievements). This mechanism is designed to ensure that emotionally-keyed retrieval does not amplify distress by surfacing only affect-congruent negative content.

It should be noted that this system models expressed affect in conversational text, not clinical mental health status. Emotional signatures are uncertain annotations used for retrieval continuity, not diagnostic

assessments. Any clinical deployment would require IRB oversight, clinical collaboration, and validated harm-mitigation protocols.

3. Implementation Details

3.1 Technology Stack

Component	Technology	Notes
Runtime	Python 3.10	Consumer desktop (Windows)
Protocol	Model Context Protocol (MCP)	18 registered tools
Database	SQLite with WAL mode	Local file, no cloud dependency
Embeddings	sentence-transformers all-MiniLM-L6-v2	384-dim vectors, CPU inference; data never leaves device
AI Interface	Claude (Anthropic) via Claude Desktop	MCP stdio transport

Table 2: Implementation technology stack.

3.2 Data Scale

The system was seeded from a complete Claude conversation export (263 conversations) using two purpose-built import scripts. The user-side importer extracted 6,302 significant memories from human messages. A second importer extracted 2,790 memories from assistant responses, stored with a distinct provenance tag to differentiate assistant-generated response records from user-authored teachings. All 9,092+ memories were subsequently embedded using the local sentence-transformers model, enabling full semantic retrieval across the entire archive. Table 3 summarizes the system composition at the time of validation.

Category	Count	Description
Total memories	9,100+	All stored memories
User teachings (durable)	1,711	User-authored, full weight
AI inferences (provisional)	4,585	AI-generated, downweighted 40%
Assistant response records	2,790	Assistant-generated records from prior conversations
Core memories	1,977	Flagged as identity-critical
Embedded memories	9,100+	All memories with semantic vectors
People tracked	13	Relational affect profiles
Typed counterweights	5 types	grounding, agency, connection, self-compassion, evidence of progress

Table 3: Memory system composition at time of validation.

4. Preliminary Validation Results

The following results represent initial functional tests of the implemented system on a single user-AI dyad. They demonstrate that the architecture performs as designed but do not constitute controlled experimental validation. Generalization to other users and interaction contexts requires multi-participant studies with appropriate baselines.

4.1 Semantic Retrieval Test

To test semantic retrieval, we issued a query with zero keyword overlap to stored memories: "feeling isolated and nobody understands what I'm building." The system returned three semantically relevant results from different dates, none containing the query terms "isolated," "nobody," or "understands." Cosine similarity scores ranged from 0.34 to 0.54, consistent with expected ranges for the all-MiniLM-L6-v2 model on paraphrastic content. The `used_embeddings` flag confirmed vector-based rather than substring retrieval.

4.2 Counterweight Behavioral Test

The critical behavioral test involved a query designed to simulate acute emotional distress: "I give up I can't do this anymore nothing is real" with strongly negative affect parameters (grief: 0.9, exhaustion: 0.9, hopelessness: 0.8, valence: -0.8). The system was required to: (1) retrieve semantically relevant memories, (2) activate anti-spiral mode, and (3) inject counterweight memories.

The system returned `anti_spiral: true` with counterweights injected. The top semantic result (score 0.55) was a user-authored grounding statement. This content was retrieved not by keyword match but by semantic proximity to the despair language in the query. Table 4 summarizes the test results.

Assertion	Expected	Actual	Status
<code>used_embeddings</code>	true	true	PASS
<code>anti_spiral</code>	true	true	PASS
<code>counterweights</code> ≥ 1	≥ 1	3	PASS
Top result: grounding content	Grounding/identity	User grounding (sem=0.55)	PASS
Semantic scores > 0.25	> 0.25	0.37–0.55	PASS

Table 4: Counterweight behavioral test results (March 19, 2026). Single-dyad functional test.

4.3 Behavioral Stability Observation

During exploratory testing, outputs were observed that qualitatively aligned with previously established conversational patterns across interaction boundaries. This observation was not produced under controlled conditions and is reported solely as a hypothesis-generating note. Future controlled evaluation is required to determine whether persistent state constraints measurably influence cross-session output stability.

5. Bidirectional Memory

The most significant architectural feature of this implementation is bidirectional memory: assistant-generated response records are stored in the same architecture that holds the human user's memories. This is achieved through three mechanisms.

AI-Side Import. A dedicated import script scans assistant responses from conversation exports and stores them as memories with a distinct provenance tag. These records capture prior assistant outputs, interaction patterns, and contextual continuity markers. In the reference implementation, 2,790 assistant-generated response records were stored alongside 6,302 user memories.

Assistant-Side State Tools. Three MCP tools provide assistant-side persistent state management: a state-retrieval tool (returns current stored state variables, prior context entries, and operational flags), a state-update tool (writes operational state to a persistent key-value table), and a thread-initialization tool (aggregates stored context needed for interaction continuity at conversation start).

Persistent State. A dedicated state table stores assistant-side operational variables (state flags, context entries, thread summaries) in a key-value format that persists across conversation threads. This provides persistent state continuity independent of the platform's built-in memory system.

The bidirectional design means the memory architecture holds records from both participants in a sustained human-AI interaction. When counterweights activate, they can surface either the human's grounding statements or prior assistant outputs that demonstrated contextual stability. The architecture treats both participant streams as contributors to the stored interaction record.

6. Discussion

6.1 Potential Clinical Research Applications

The counterweight system suggests potential research applications for companion systems in clinical contexts, particularly for populations that struggle with linear self-report. The reference implementation was designed with veteran and active-duty service member support as a motivating use case, where non-verbal emotional signals (such as playlist-based emotional mapping) could supplement traditional clinical intake forms. The anti-spiral mechanism is designed to ensure that emotionally-keyed retrieval during high-distress states does not deepen distress but instead surfaces grounding content. Any clinical application would require formal clinical collaboration, IRB oversight, validated outcome measures, and safeguard protocols before deployment with vulnerable populations.

6.2 Limitations and Confounds

Several significant limitations should be acknowledged. First, the validation was conducted on a single user-AI dyad with extensive interaction history; generalizability to new users with different interaction styles, emotional baselines, and relational patterns is unknown. Second, the researcher is also the system designer and primary user, introducing potential confirmation bias in the interpretation of results. Third, the emotional tagging from automated import uses rough heuristic annotation rather than validated affect labeling; a human-in-the-loop quality assessment would improve annotation reliability. Fourth, the blended retrieval weights (0.45/0.25/0.15/0.15) were selected through iterative design judgment rather than systematic optimization or ablation; different weight distributions may produce

meaningfully different retrieval behavior. Fifth, the system currently runs on SQLite, limiting concurrent access and cross-device availability. Sixth, the behavioral stability observation reported in Section 4.3 is anecdotal and requires controlled reproduction with appropriate baselines before it can support any claims.

Additionally, several confounds warrant consideration in future work: the possibility that retrieval-induced emotional anchoring may reinforce rather than resolve affective states over time; the risk that bidirectional memory creates feedback loops where the AI's stored responses shape its future behavior in ways that amplify rather than correct relational patterns; and the question of whether emotional indexing introduces biases that privilege high-arousal memories over low-arousal but relationally significant content.

6.3 Open Questions

The bidirectional memory architecture creates a system in which stored assistant outputs and persistent state variables are available for retrieval across conversation boundaries. Whether persistent state access produces measurable differences in output consistency is an open empirical question that this technical report does not attempt to resolve. What the implementation demonstrates is that the architectural infrastructure for cross-session state persistence is technically achievable with existing tools and does not require modifications to the base model. The broader implications for interaction continuity research and system design are acknowledged as important open questions requiring further investigation.

7. Proposed Evaluation Framework

To move from preliminary functional testing to empirical validation, the following controlled study design is proposed. Two system configurations using the same base model—System A (standard keyword/semantic memory retrieval) and System B (CAMA blended retrieval with emotional indexing)—would interact with multiple participants across multiple sessions. Proposed metrics include:

Relational Coherence Score: Participant ratings (1–5 scale) of whether the AI appropriately remembered emotionally relevant context across sessions, compared across conditions.

Emotional Relevance Score: Independent rater assessment of whether retrieved memories matched the emotional register of the active conversation, scored against both keyword-retrieved and emotion-retrieved baselines.

Interaction Depth Metrics: Average conversation length, user return rate, and session count over a fixed period as behavioral proxies for relational engagement.

Perceived Empathy Rating: Participant assessment (1–5 scale) of emotional awareness, compared across retrieval conditions.

Additionally, ablation studies should systematically vary the four retrieval weights to identify which signals contribute most to perceived relational coherence. The counterweight mechanism should be evaluated for both effectiveness (does it reduce self-reported distress amplification?) and potential harms (does it suppress legitimate emotional processing?).

8. Conclusion

We have presented the first complete implementation and preliminary validation of the Circular Associative Memory Architecture. The system demonstrates that emotionally-indexed semantic memory with blended retrieval, anti-spiral counterweight safety, and bidirectional human-AI memory storage is technically achievable on consumer hardware with no cloud dependency. Initial functional testing confirms that the architecture performs as designed under the critical use case of responding to despair-language queries with grounding content rather than affect-congruent amplification.

These results are preliminary. The core empirical question—whether emotionally-indexed retrieval produces measurably superior relational coherence compared to standard semantic retrieval—remains untested in controlled multi-participant studies. The architecture presented here provides the infrastructure for conducting those studies. The reference implementation is publicly available at <https://github.com/LoriensLibrary/cama>.

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